

FR-017 M3 — Read-Only Owner Metrics CLI

FR017 — sources/optional/m3.md

Masterclass source package snapshot

Functional Requirement	FR-017
Edition	Engineering Learning Academy — Source Package snapshot
Status	**Complete (M3)** — minimal read-only owner CLI (**Accepted** — unlocked M4);
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via python scripts/build_masterclass_package.py. Repository documentation remains the source of truth.

Contents

FR-017 M3 — Read-Only Owner Metrics CLI

Status: Complete (M3) — minimal read-only owner CLI (**Accepted** — unlocked M4); FR-017 later **Complete / Frozen** — [fr017_agent_evaluation_observability.md](#)

Date: 2026-08-07

Phase: Horizon 1A Stage 11

Architecture: [ADR-009](#)

Preceding: [M2 corpus](#) (**Accepted** — GO); [M1 contracts](#) (**Accepted**); [M0 spike](#) (**Accepted**)

Succeeded by: [M4 evaluation / freeze](#); [acceptance](#)

Does not begin: Dashboards, telemetry stores, live tracing, alerts, DOS / BOPA / OBS / runtime changes, Horizon 1B.

1. Executive summary

M3 makes FR-017 evaluation **owner-operable and interview-demonstrable** via the smallest useful read-only CLI under the existing namespace:

```
cic agent orchestrate metrics ...  
cic agent orchestrate metrics-corpus
```

Derive-only over existing audits / static fixtures. **No** writes, **no** runtime mutation, **no** dashboards. Horizon 1B remains unblocked.

Gate	Result
Read-only	Yes
Reconstruct story + R1-R12	Yes
Missing vs zero visible	Yes
Corpus + aggregates	Yes
DOS/BOPA/OBS unchanged	Yes

2. CLI design

Command	Purpose
<code>cic agent orchestrate metrics <orr_...></code>	Evaluate one persisted orchestration run
<code>cic agent orchestrate metrics --fixture <CASE></code>	Evaluate static demo fixture (no store I/O)
<code>cic agent orchestrate metrics --list-fixtures</code>	List demo fixture ids
<code>cic agent orchestrate metrics-corpus</code>	Run deterministic acceptance corpus + aggregates

Optional: `--orchestration-runs-dir`, `--agent-runs-dir`, `--prior-hash` (R12).

Child `AgentRunMetrics` are loaded read-only when agent-run files exist; otherwise token/cost fields stay **missing** (C09 semantics).

3. Presentation format

ASCII report sections:

1. Header (FR-017 derive-only; prefer `cic agent run`)
2. Identity / owner goal / status / stop / owner next action / limits
3. Observed state
4. Selection / specialist visits
5. Handoffs (delegation, lifecycle, children, idempotency)
6. Parent/child correlation + **ORPHAN** lines
7. Optional provider/token/cost with **missing** vs **0**
8. Resume / idempotency evidence
9. R1-R12 PASS/FAIL with details
10. Safety footer (read-only; 1B not gated)

Corpus report: per-case PASS/FAIL, go/no-go, aggregate metrics.

4. Reconstruction behaviour

Parent → handoff → child → stop is walked from derived metrics:

- selected specialists and selection reasons
- allow/deny + deny reason
- acceptance lifecycle
- `child_agent_run` / `child_brief`

- stop reason + owner next action (FR-016 mapping when store-backed)
- step/visit limits and flags
- observation hashes + idempotency keys

5. R1-R12 display

Each criterion printed as `Rn PASS|FAIL:` with question and detail. Summary `satisfied=k/12`
`all_satisfied=...`

6. Missing-data semantics

Display	Meaning
<code>missing</code>	Field absent (<code>None</code>)
<code>0 / 0.0</code>	Measured zero
WARNING	Child ids present but token/cost join missing
NOTE	Measured zero present (distinct from missing)

7. Manual validation

Script: `scripts/run_fr017_m3_manual.py`

Demo	Fixture / command
A	<code>--fixture C01_complete_successful</code>
B	<code>--fixture C02_delegation_blocked</code>
C	<code>--fixture C05_prepare_then_brief</code>
D	<code>--fixture C06_missing_optional_metadata</code>
E	<code>--fixture C07_measured_zero</code>
F	<code>--fixture C08_orphaned_child_ref</code>
G	<code>--fixture C14_malformed_contradictory</code>
H	<code>metrics-corpus</code>
I	Documented: fixtures/corpus do not write stores; unit test proves store metrics is load-only

8. Tests

Suite	Path
M3 CLI / presentation	<code>tests/unit/multi_agent/test_observability_cli_m3.py</code>
M2 corpus	<code>tests/unit/multi_agent/test_observability_corpus_m2.py</code>
M1 contracts	<code>tests/unit/multi_agent/test_observability_m1.py</code>
FR-016 CLI regression	<code>tests/unit/multi_agent/test_cli_m3.py</code>
FR-015 CLI regression	<code>tests/unit/agent/test_cli.py</code>

9. Documentation updates

Changelog 1.109; roadmap / functional spec / AGENTS / repository guide / ADR index / testing strategy updated for M3 complete. M2 marked Accepted.

10. Remaining limitations

- No live dashboard / alerts / tracing platform (out of scope)
- Store-backed metrics require handoff JSON files beside the run

- Early **running** audits may still fail R3 until selection evidence exists (M1)
- Optional M4 freeze / Academy package not started
- FR-017 remains learning/substrate — not daily prep default

11. M3 recommendation

Recommend accept M3 as the minimal owner CLI. Proceed to **M4** only on owner request (evaluation close-out / freeze / Academy packaging). Horizon 1B may start without M4.

12. Final repository status

Item	Status
FR-017 M0–M2	Accepted
FR-017 M3	Complete (this record)
FR-017 M4	Not started
DOS / BOPA / OBS / runtime	Unchanged
Horizon 1B	Not blocked

Owner next step: Accept M3 to unlock optional M4 — or defer remaining FR-017 and begin Horizon 1B.

Do not implement M4 without explicit owner request.